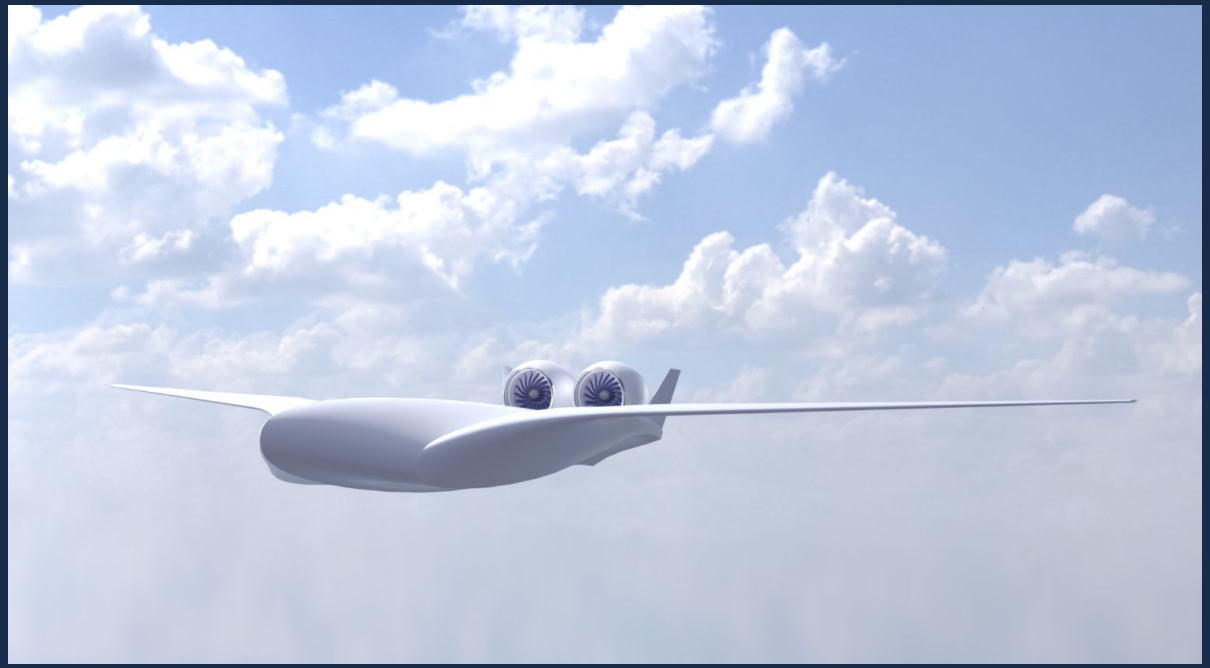
UC San Diego

Conceptual Design Review

MAE 155A Group 2 - Ultra Fuel-Efficient 250 Passenger Aircraft



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Executive Summary

Fuel burn per passenger has become stagnant

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Marginal gains left in Tube-and-Wing optimization

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Need for innovative aircraft architectures

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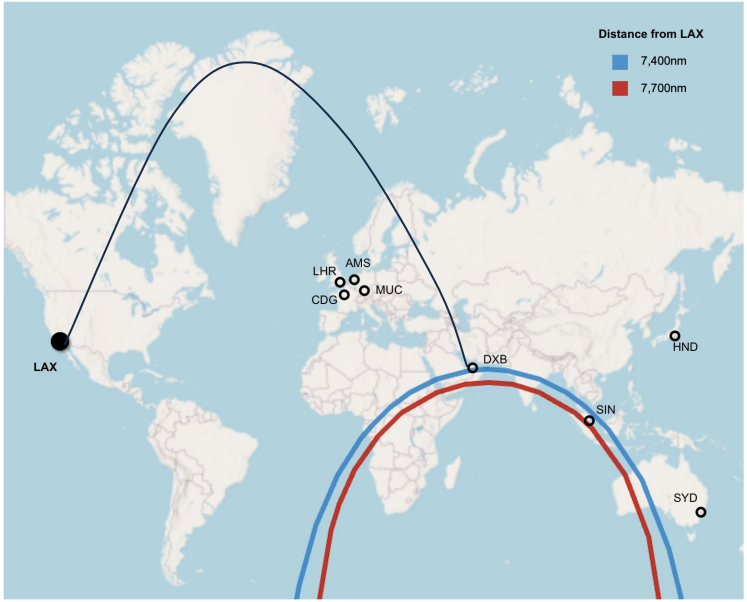
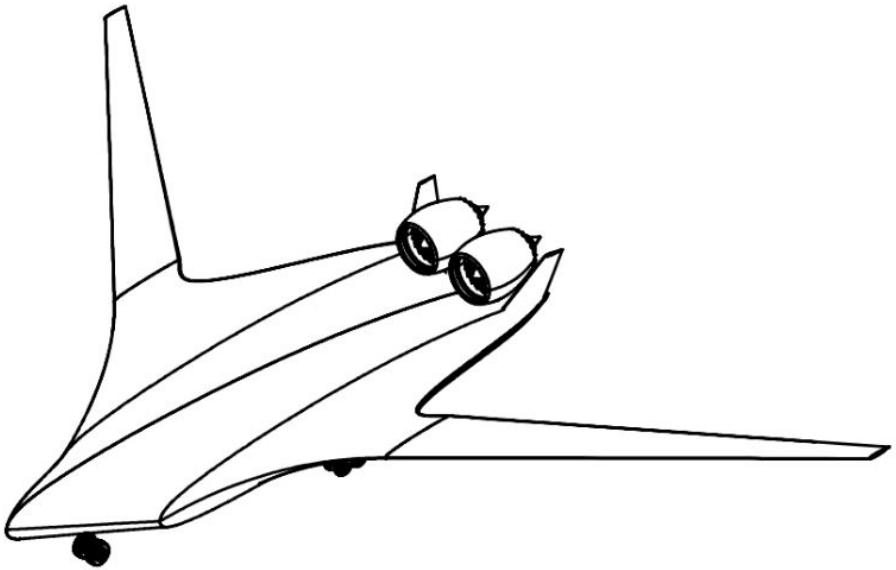
Blended-Wing-Body



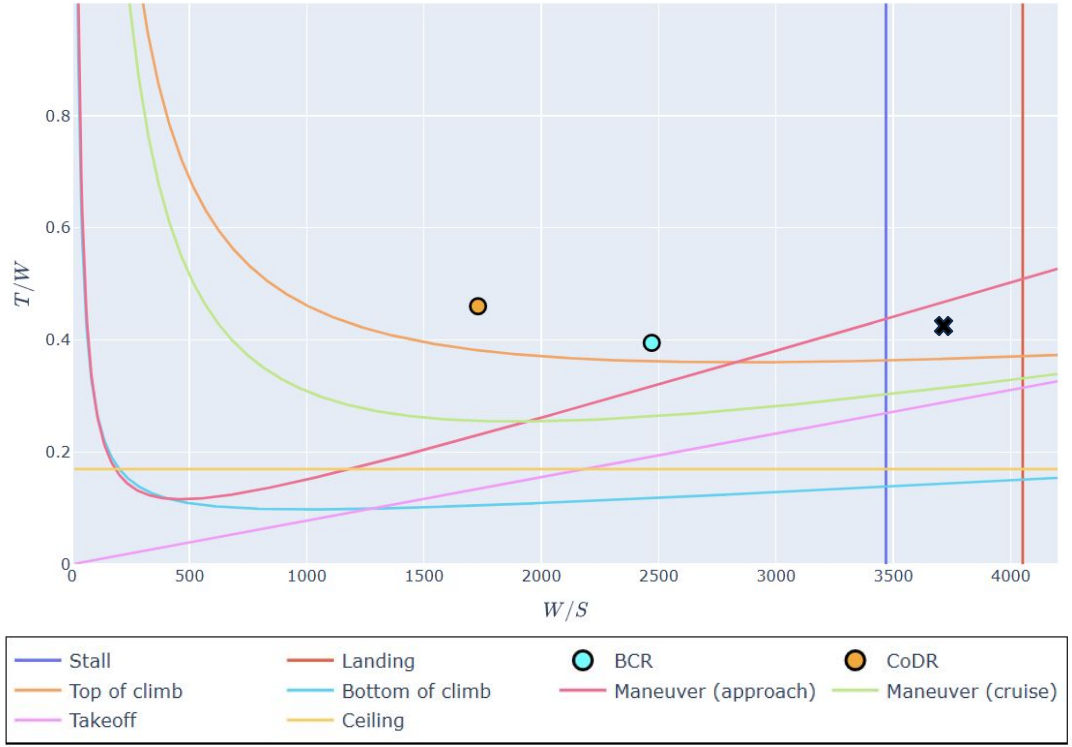
Range [nmi]	Cruise Mach	Altitude [ft]	Capacity [pax]	Wingspan [m]	Cruise L/D
7400	0.85	40,000	250	64.67	25.30

Main Design Choices

- Fuselage as main lifting body
- Non-cylindrical cabin section
- Aft mounted high-bypass turbofan engines
- Twin vertical stabilizers

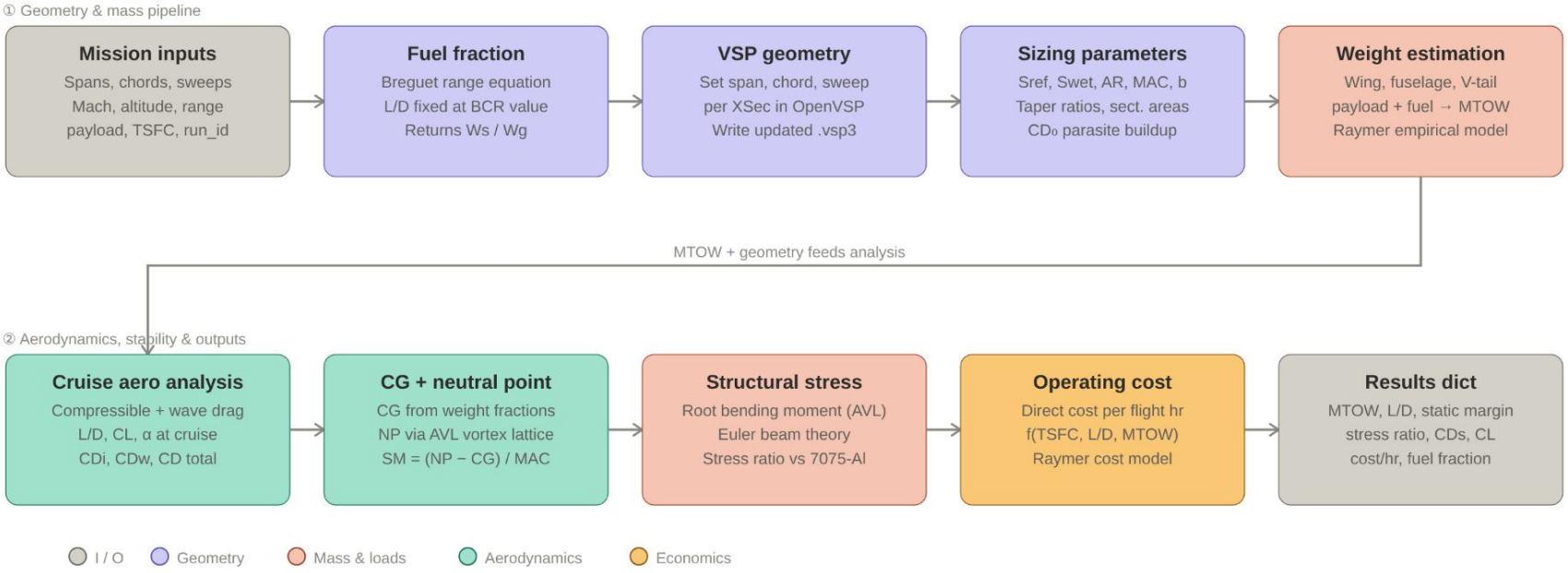


Aircraft Sizing

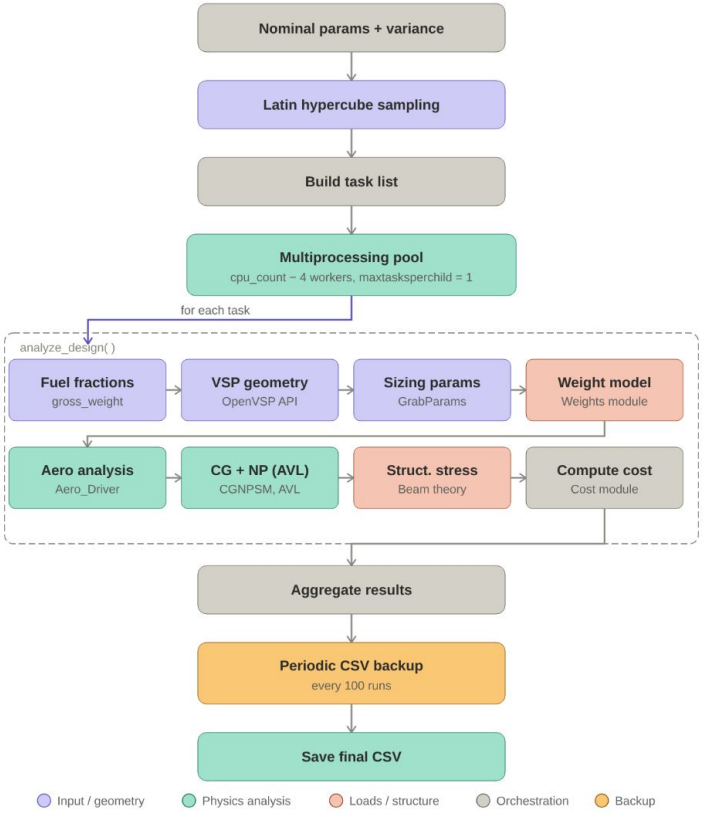


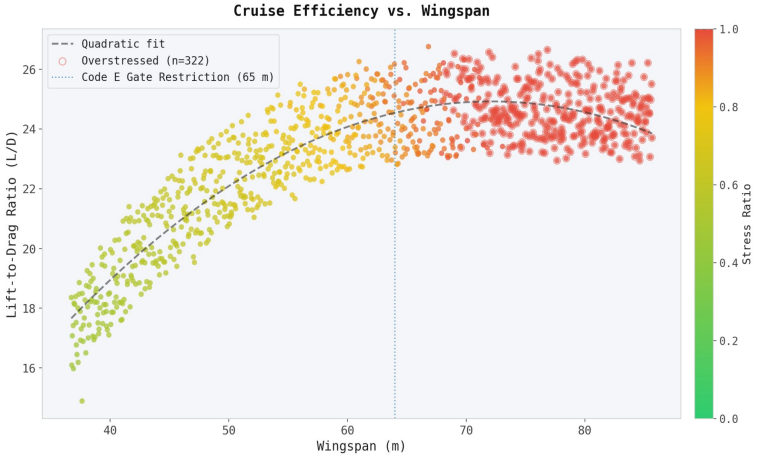
	Initial	BCR	CoDR
W_g [N]	1.71e+6	1.67e+6	1.44e+6
W_p/W_g	.23	.23	0.27
W_e/W_g	0.41	0.41	0.42
W_s/W_g	0.36	0.36	0.31
W/S [N/m^2]	3680	2470	1730
T/W	0.42	0.39	0.46

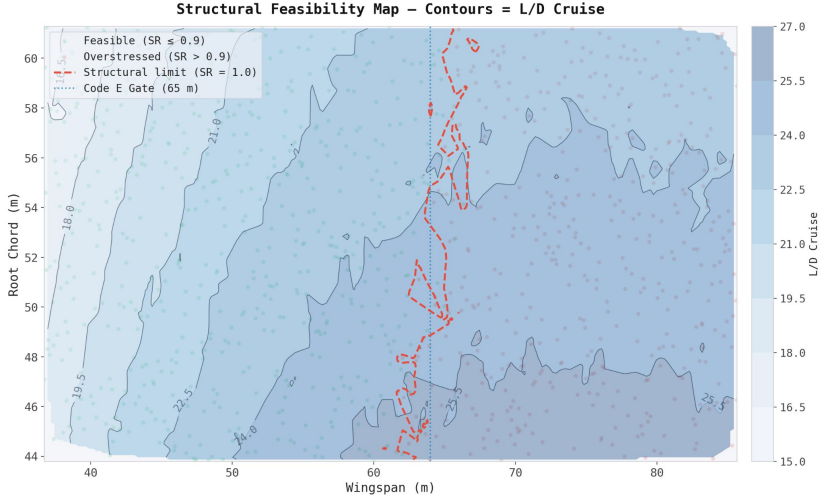
Master Script



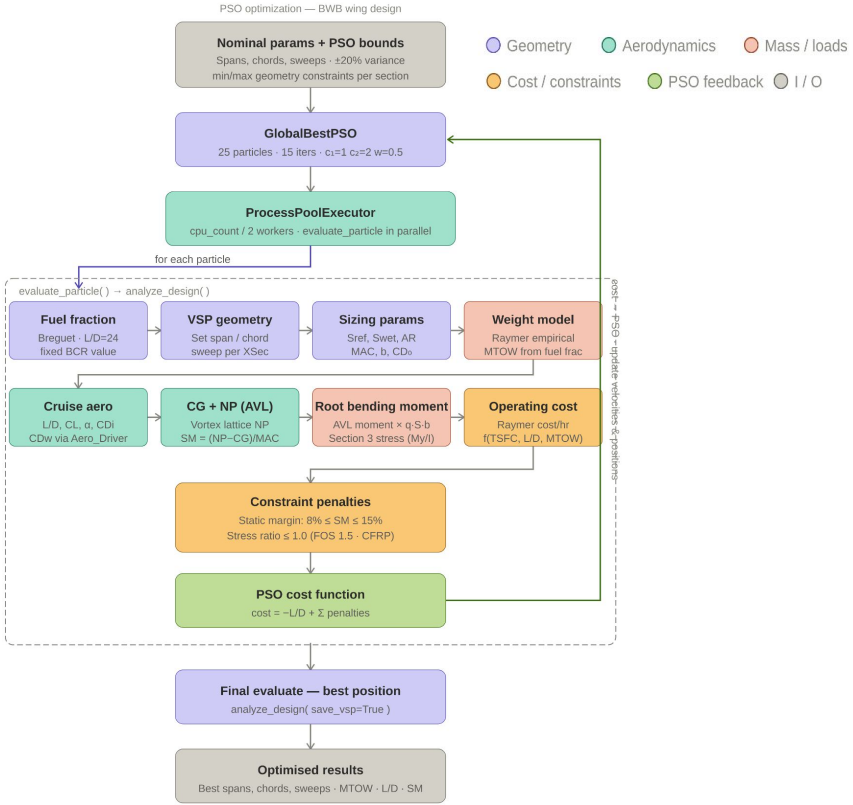
Monte Carlo


Nominal params + variance
Latin hypercube sampling
Build task list
Multiprocessing pool
cpu_count − 4 workers, maxtasksperchild = 1
for each task
analyze_design( )
Fuel fractions
gross_weight
VSP geometry
OpenVSP API
Sizing params
GrabParams
Weight model
Weights module
Aero analysis
Aero_Driver
CG + NP (AVL)
CGNPSM, AVL
Struct. stress
Beam theory
Compute cost
Cost module
Aggregate results
Periodic CSV backup
every 100 runs
Save final CSV
Input / geometry
Physics analysis
Loads / structure
Orchestration
Backup


Cruise Efficiency vs. Wingspan
Quadratic fit
Overstressed (n=322)
Code E Gate Restriction (65 m)
Lift-to-Drag Ratio (L/D)
Wingspan (m)
Stress Ratio


Structural Feasibility Map — Contours = L/D Cruise
Feasible (SR ≤ 0.9)
Overstressed (SR > 0.9)
Structural limit (SR = 1.0)
Code E Gate (65 m)
Root Chord (m)
Wingspan (m)
L/D Cruise

Gradient Free Optimization



Design variables:

$$\mathbf{x} = [\mathbf{AR}, \mathbf{S}, \boldsymbol{\lambda}, \boldsymbol{\Lambda}]; \quad \mathbf{AR}, \mathbf{S}, \boldsymbol{\lambda}, \boldsymbol{\Lambda} \in \mathbf{R}^3$$

Objective:

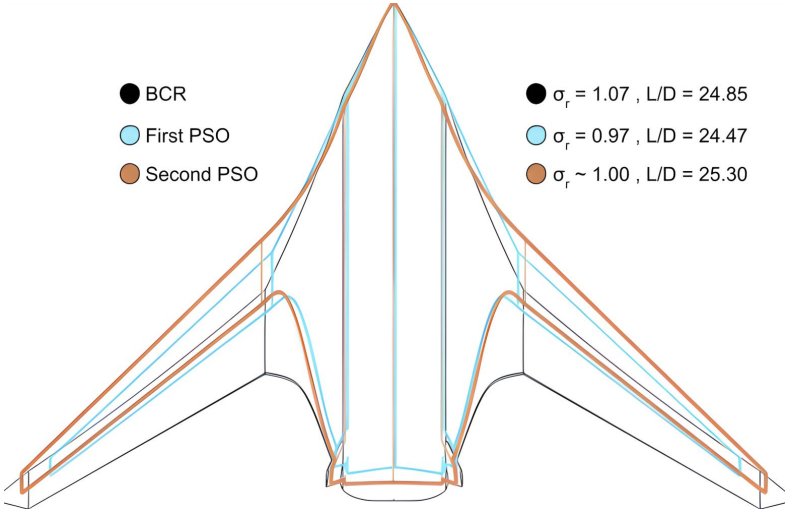
$$\max_{\mathbf{x}} \frac{L}{D}(\mathbf{x})$$

Constraints:

$$\sigma(\mathbf{x}) \leq \sigma_{\text{allow}}, \quad SM(\mathbf{x}) \geq 0,$$

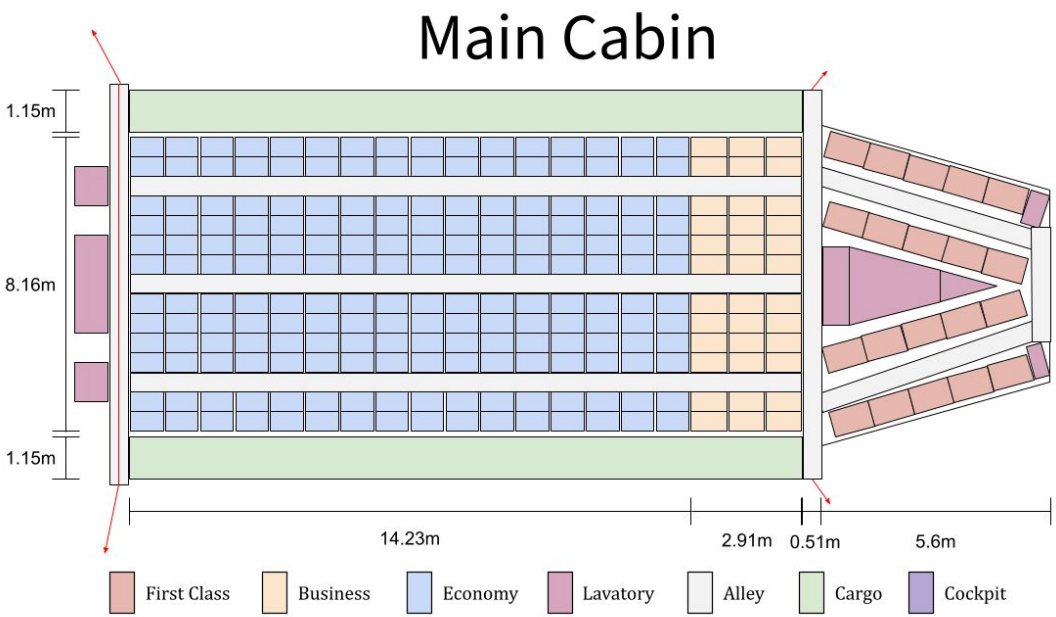
Bounds:

$$[\mathbf{AR}_{\min}, \mathbf{S}_{\min}, \boldsymbol{\lambda}_{\min}, \boldsymbol{\Lambda}_{\min}] \leq \mathbf{x} \leq [\mathbf{AR}_{\max}, \mathbf{S}_{\max}, \boldsymbol{\lambda}_{\max}, \boldsymbol{\Lambda}_{\max}]$$

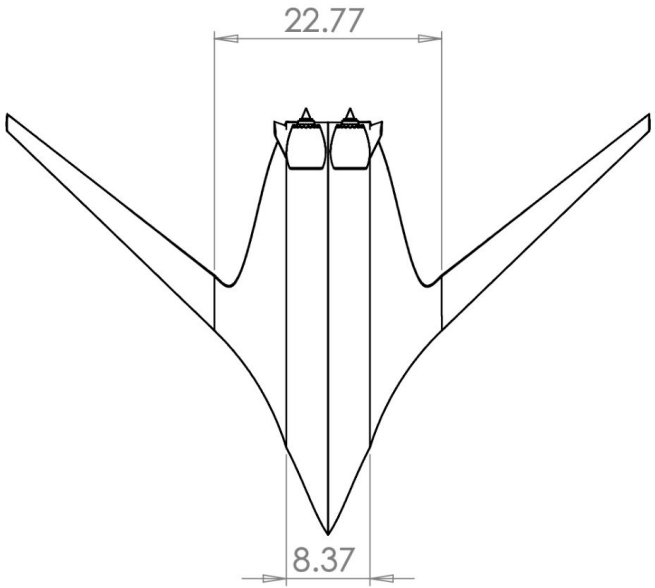


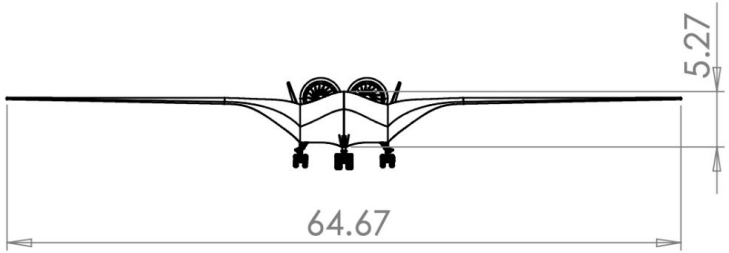
3-Class Layout

- Crew space below first class
- 2-4-4-2 seating arrangement
  - 20 first class seats
  - 3 business class rows
  - 16 economy class rows
- Premium amenities



Aerodynamics





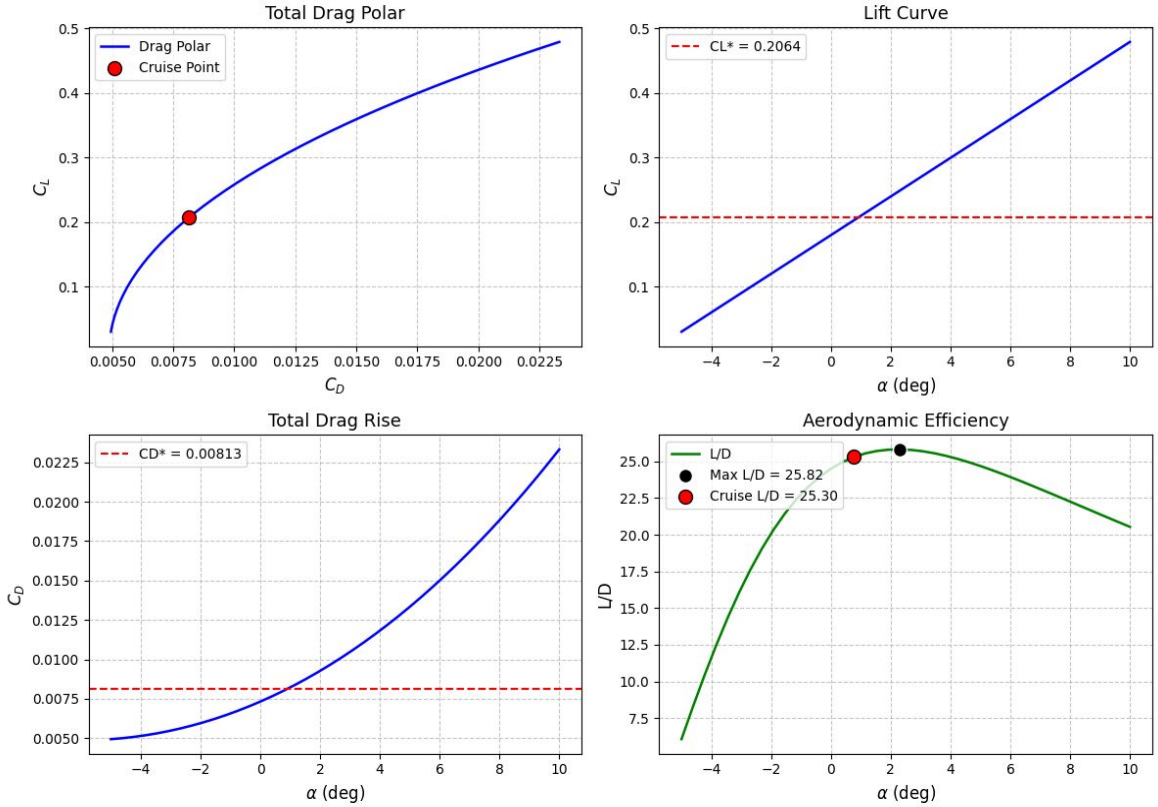
Wing Design:

Aspect Ratio	5.03
Wingspan (m)	64.67
Fuselage Height (m)	5.27
Fuselage Length (m)	41.19

Airfoil Selection:

	Center Body	Outer Wing
Airfoil	NACA 25111	SC(2)-0410
Sweep (°)	65	44
t/c (%)	11	10

Cruise L/D of 25.3



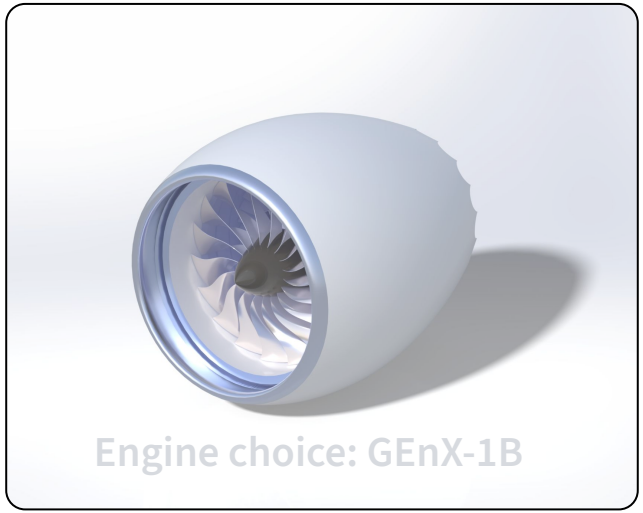
Cruise C_L	0.21
Cruise AoA (°)	0.896
$C_{D,0}$	0.0049
$C_{D,induced}$	0.0029
$C_{D,wave}$	0.0003
C_D	0.0081
L/D	25.3
$dC_L/d\alpha$ (π)	1.71

Ultra-high bypass turbofan engines

TSFC (1/s)	Thrust (kN)	Mass flow (kg/s)	Rotor Blades	Dry Mass (kg)	Bypass Ratio
1.42e−5	330	1160	18	6147	9.1

Configuration:

- Dual-engine
- High-Bypass Ratio Turbofan
- Rear-mounted (upper airframe surface)
- Maximize Boundary Layer Ingestion
- Minimize Cabin Noise



MTOW dictated by fuel and payload

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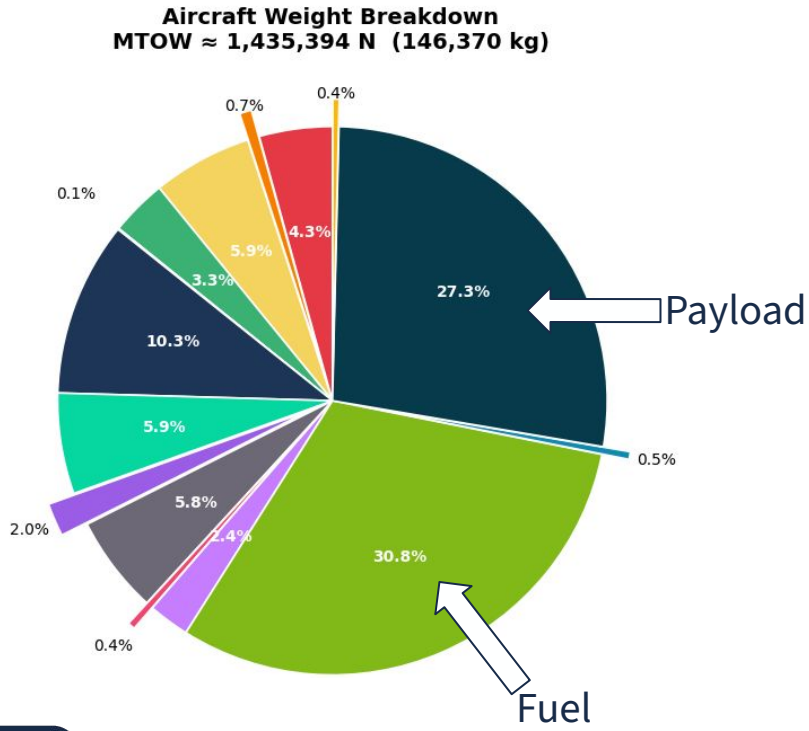

Section	Weight (N)
Fuel	4.424e5
Payload	3.920e5
Wing + Fuselage	1.458e5
Engine	1.477e5
Flight Controls	8.520e4
Landing Gear	4.877e4
Miscellaneous	1.731e5
Total	1.435e6

The static margin shifts forward in flight

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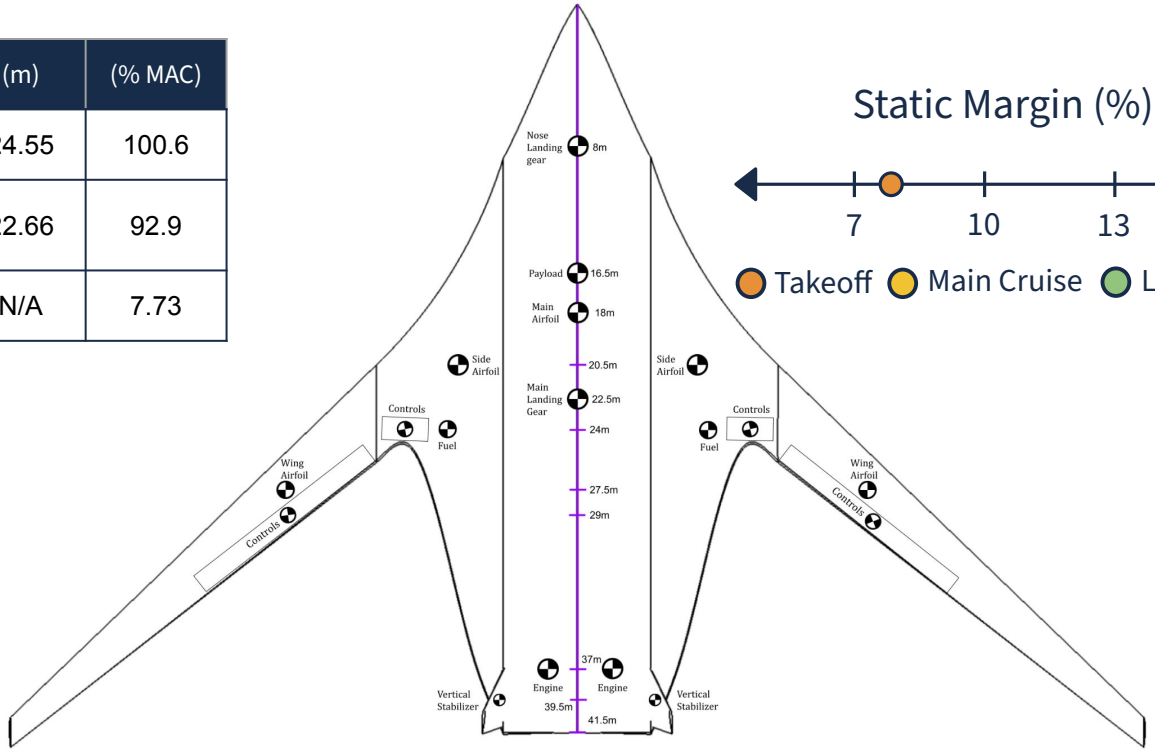
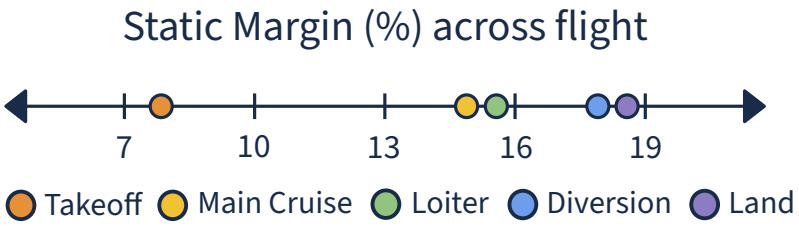
Alex

Parameter	(m)	(% MAC)
Neutral Point	24.55	100.6
Center Of Gravity	22.66	92.9
Static Margin	N/A	7.73





Dynamic Stability & Controls

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Fly By Wire (FBW) Implementation:

- 3 Independent control systems
- Triplex flight control computer
- Master control system for signal integration
- Robust modeling and flight testing

Longitudinal Stability Parameters:

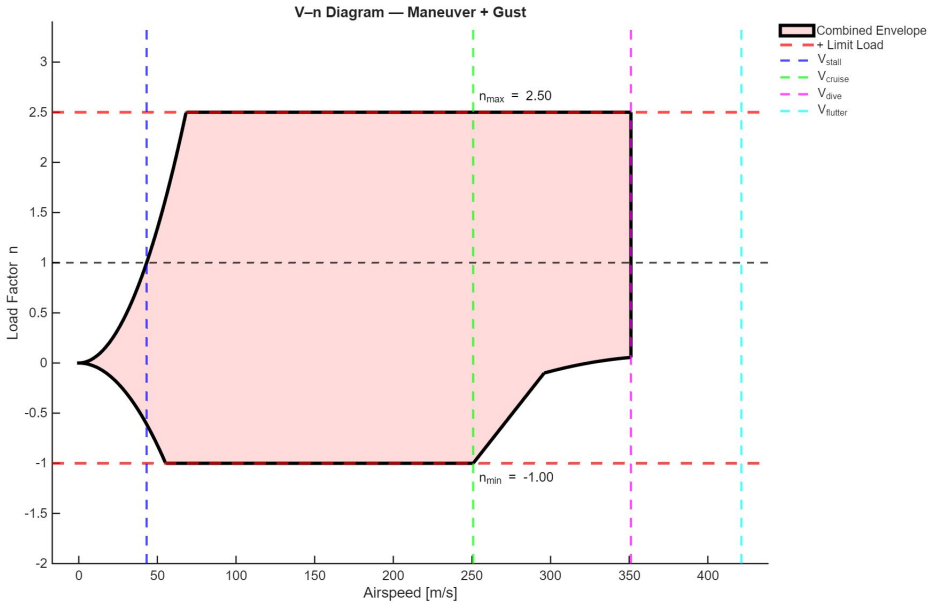
Mode	Damping Ratio	Time to Double [s]	Acceptance Level
Short Period	0.303	—	2
Phugoid Period	—0.0314	600	3

Lateral Stability Parameters:

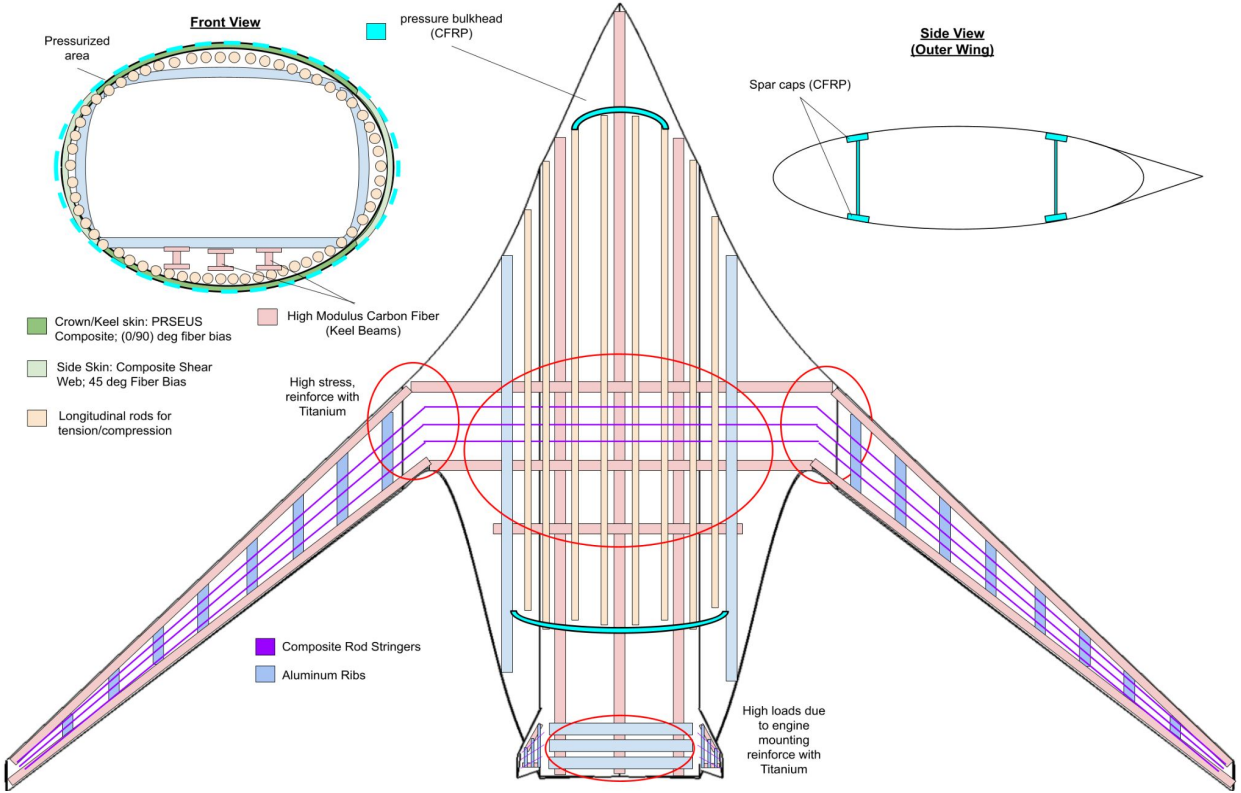
Mode	Damping Ratio	Time to Double/half [s]	Acceptance Level
Roll Subsidence	1	0.52 (half)	2
Spiral Divergence	—1	4.06 (half)	3
Dutch Roll	0.043	8.36 (double)	2

Structures and Materials

- Primary lifting structures
  - Carbon fiber reinforced polymer (CFRP)
- Beam Structures
  - High Modulus Carbon Fiber
- Wing and Vertical tail ribs
  - Aluminum
- Engine mounts, Wing-body transition
  - Titanium
- Landing gear components
  - Steel
- Metallic reinforcements
  - Titanium, Aluminum

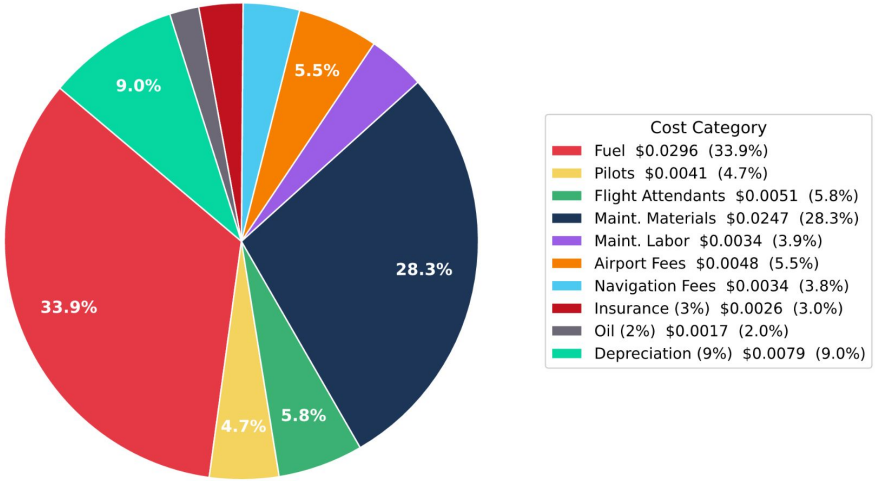


Structural Layout



Direct Operating Cost

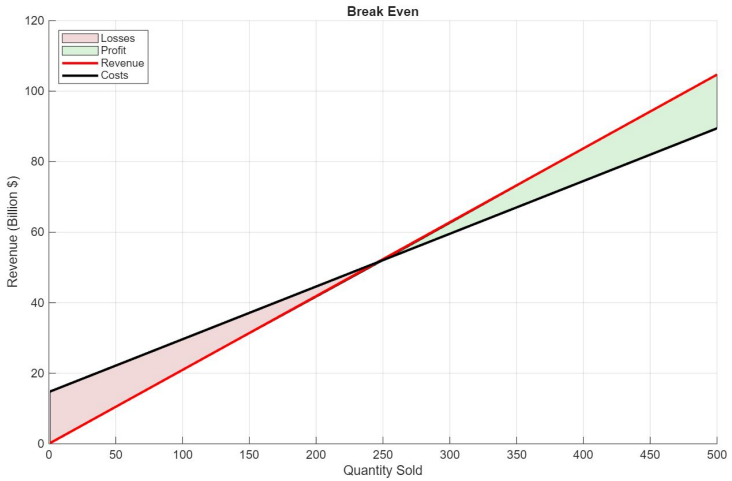
- Direct operating cost: $12,237 per flight hour
- Cost per Seat Mile: $0.0873
- Flight From Los Angeles (LAX) to Dubai (DXB)
  - Operational for flight route: ~$187,000

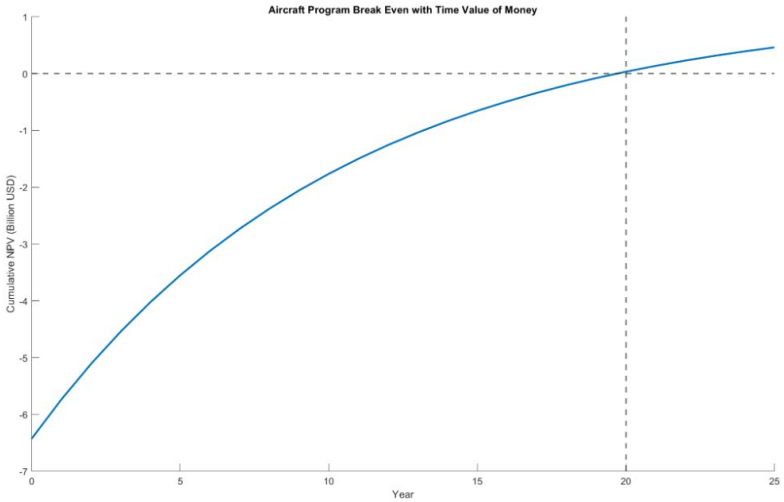


Break Even Analysis

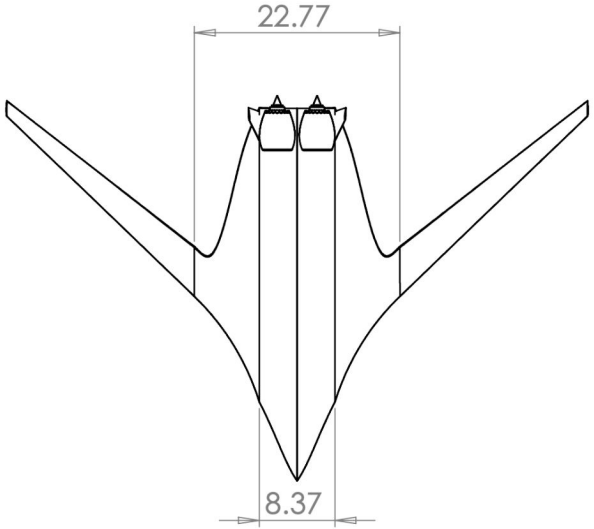
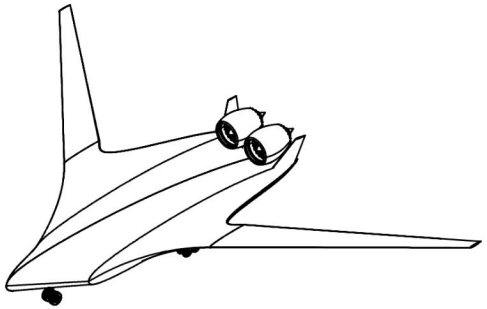
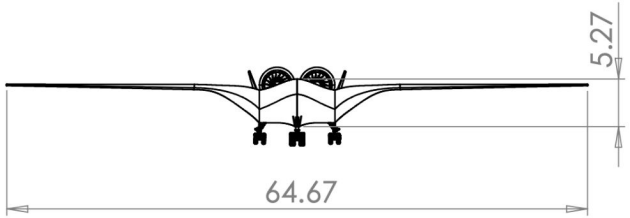
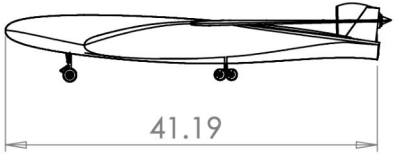
Break Even:

- 249 Aircraft Sold
- $51.8 Billion In Revenue
- Cost of $149.5 Million Per A/C
- Sale price of $209.5 Million Per A/C





Time Value of Money

- ~20 Years to break even
- 6.5 billion into development program

Thank you.